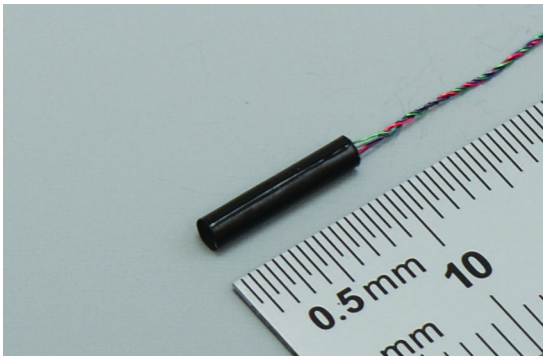
Aurora Mini 6DOF Sensor, 1.8 x 9 mm

Introduction

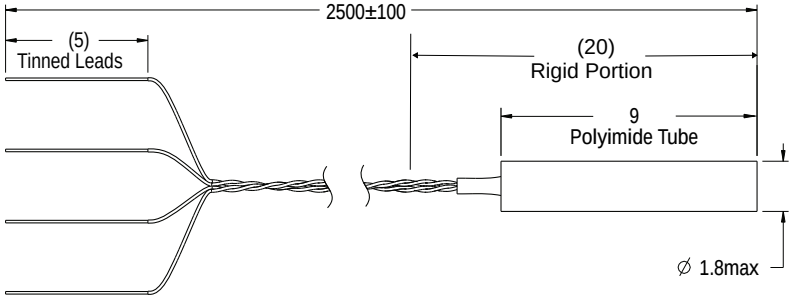
This data sheet is applicable to the Aurora® Mini 6DOF Sensor, 1.8 x 9 mm, Part Number 610029. For further information on your Aurora System, refer to the documentation that was delivered with the system.

The Aurora 6DOF Sensor consists of two five degrees of freedom (5DOF) sensors, fixed relative to each other and secured in a polyimide tube. It provides position and orientation data in six degrees of freedom (6DOF). Its small size allows it to be integrated into applications where space restriction is a design factor.

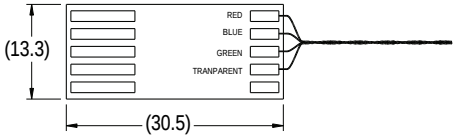


Description

The Aurora 6DOF Sensor consists of two 5DOF sensors, each mounted on a printed circuit board (pcb). Each sensor has a resistance of ~95 Ω. The two pcbs are fixed together, forming a sub-assembly of two sensors that are fixed relative to each other. The sub-assembly is housed inside a polyimide tube. A dimension reference drawing is shown below.



Sensor is supplied with test adapter (pcb)
soldered as below, at the sole discretion of NDI.



All dimensions in mm

To integrate the Aurora 6DOF Sensor in a tool, you must use a generic tool definition file (610029.rom). This file is located at the Aurora Software section of the NDI Support Site at:

http://support.ndigital.com

Program this file into the tool's Serial Read Only Memory (SROM) device, using NDI's 6D Architect (6DA) (must be version 3 or later) software or load it directly in the memory of the Aurora System. The file contains sensor characterization information and may be further customized by 6DA to include additional data, for example, offsets and tool serial numbers. (Do not change "Tool Class" (from "6DOF Tool") or the relative positions/orientations of the two sensors.)

The Aurora 6DOF Sensor is supplied with flying leads, arranged as two twisted pairs and with a total length of approximately 2.5 m. To be consistent with the tool definition file, the connections between the Aurora 6DOF Sensor and the (10 pin) tool connector must be made as follows:

Tool Connector Pin #	Wire Colour		Connection
1	Red	Sensor 1	Start of sensor 1
2	Blue		End of sensor 1
3	Green	Sensor 2	Start of sensor 2
4	Transparent		End of sensor 2

Sensor Wiring

Before connecting any wiring to the sensors, read the following:

1.Take care to avoid damage to the sensor lead wires when handling and mounting the sensor.

2.Securely mount the sensors within the tool body. If a sensor moves out of position, it will affect the accuracy of the tool.

3.Make sure the sensors are electrically isolated from all metallic or conductive materials.

4.Design the tool body such that the sensor does not come into contact with moisture. If a sensor comes into contact with moisture, it may affect the accuracy of the tool.

Accuracy

For information on tracking accuracy, contact NDI technical support, at the office nearest your location. See below for contact details.

Disclaimer

Aurora sensors are not approved for medical use. Testing, certification and validation should be completed by the original equipment manufacturer or the end user prior to use in any medical application, or any application involving living humans. Please contact NDI for details. Users must determine the suitability of Aurora sensors for their own applications.

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DDDS-610-029 Revision 4.